OFFICIAL ARCHITECTURE SPECIFICATION

EduPulse — System Architecture & Technical Specification

Digital Educational Data Management Platform with Dynamic Cryptographic Attendance Automation, Anti-Proxy Guard & AI Biometric Liveness

DOCUMENT VERSION	BACKEND STACK	FRONTEND STACK	DATABASE
v1.0.0 (Production)	Node.js / Express 5 / Socket.io	React 19 / Vite 8 / Tailwind v4	PostgreSQL 18 / Prisma 6

1. Executive Summary & Problem Formulation

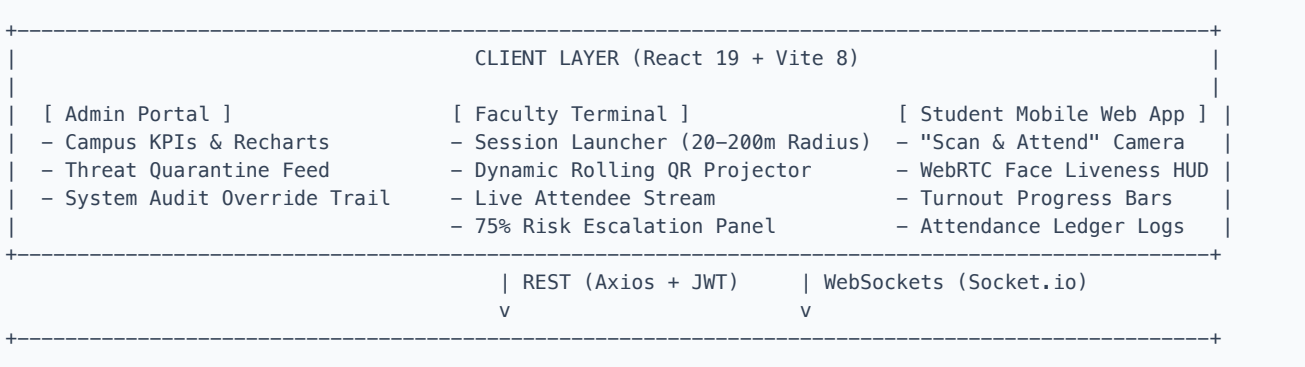
Educational institutions worldwide suffer substantial administrative overhead and academic integrity breaches due to **proxy attendance fraud**. Traditional attendance paradigms present critical points of failure:

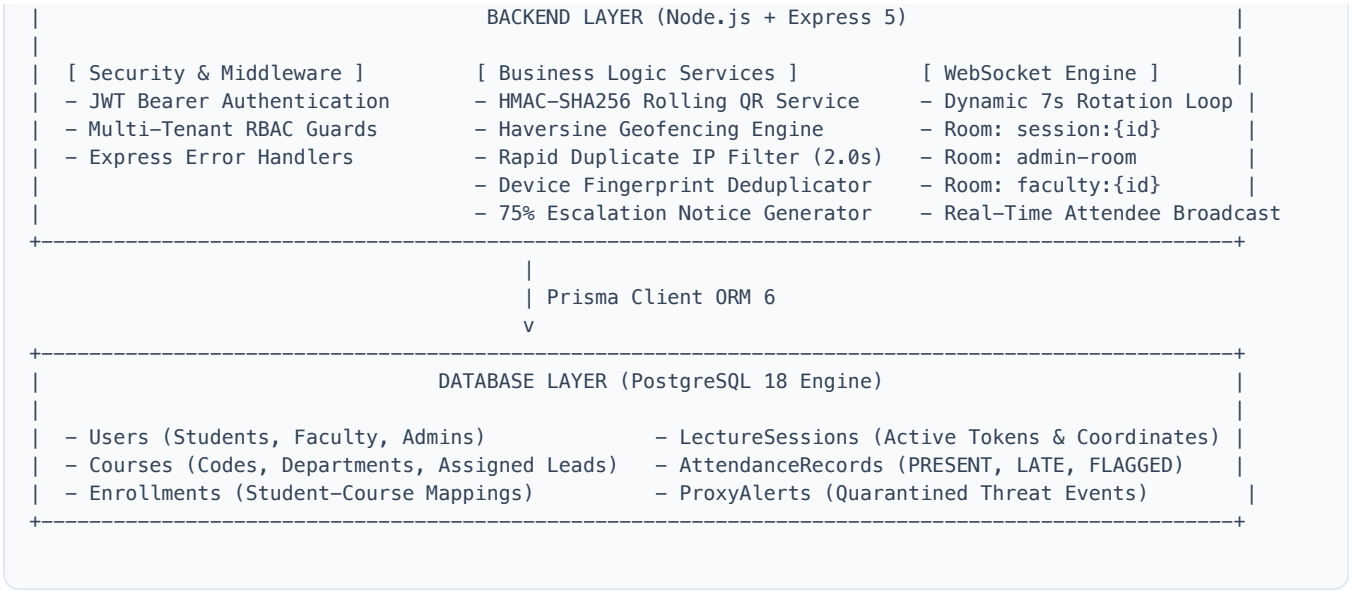
- **Paper Sign-in Sheets:** Easily forged by students signing on behalf of absent peers; manual tallying wastes 10–15% of class lecture time.
- **Static QR Code Placards:** Students photograph the QR code using a smartphone and broadcast it via WhatsApp or Telegram groups to students outside campus grounds.
- **RFID & Dedicated Biometric Hardware:** Prohibitive per-classroom installation and maintenance costs; queues cause hallway congestion; susceptible to physical card relaying.
- **Delayed Debarment Detection:** Students falling below the statutory **75% regulatory attendance threshold** are typically only identified at end-of-term auditing, preventing early remediation.

EduPulse provides a zero-hardware-fixture, full-stack educational data management platform. It combines dynamic rolling cryptographic tokens (refreshed every 7 seconds via WebSockets), spherical GPS geofencing, duplicate IP network anomaly analysis, hardware device fingerprinting, and browser-native AI facial recognition to render proxy attempts virtually impossible.

2. End-to-End System Architecture

EduPulse employs an event-driven, decoupled client-server architecture. Real-time updates occur via bidirectional WebSockets while persistent data operations run over a secured RESTful API backed by PostgreSQL and Prisma ORM.




BACKEND LAYER (Node.js + Express 5)
[Security & Middleware]
- JWT Bearer Authentication
- Multi-Tenant RBAC Guards
- Express Error Handlers
[Business Logic Services]
- HMAC-SHA256 Rolling QR Service
- Haversine Geofencing Engine
- Rapid Duplicate IP Filter (2.0s)
- Device Fingerprint Deduplicator
- 75% Escalation Notice Generator
[WebSocket Engine]
- Dynamic 7s Rotation Loop
- Room: session:{id}
- Room: admin-room
- Room: faculty:{id}
- Real-Time Attendee Broadcast
Prisma Client ORM 6
DATABASE LAYER (PostgreSQL 18 Engine)
- Users (Students, Faculty, Admins)
- Courses (Codes, Departments, Assigned Leads)
- Enrollments (Student-Course Mappings)
- LectureSessions (Active Tokens & Coordinates)
- AttendanceRecords (PRESENT, LATE, FLAGGED)
- ProxyAlerts (Quarantined Threat Events)

3. The 4-Tier Anti-Proxy Threat Quarantine Engine

When a student submits attendance, the request must successfully pass four distinct validation layers before being registered as `PRESENT`. If any layer fails, the transaction is immediately quarantined as a `FLAGGED_PROXY` and broadcast to the administrative security feed.

Tier 1 Dynamic Cryptographic Rolling QR

The faculty terminal displays an SVG QR code that rotates every **7 seconds** over WebSockets. The payload encapsulates: `sessionId`, a millisecond `timeBucket`, and a cryptographic `HMAC-SHA256` signature.

Tokens older than **15 seconds** or with mismatched hashes are rejected. This completely neutralizes WhatsApp/Telegram screenshot sharing.

Tier 2 Haversine GPS Geofencing

During scanning, the student's mobile browser acquires device GPS coordinates via `navigator.geolocation`. The backend computes the spherical great-circle distance to the classroom anchor coordinates.

If the distance exceeds the professor's configured radius (default **50 meters**), the attempt is quarantined as a `GEOLOCATION_MISMATCH`.

Tier 3 Rapid-Burst Duplicate IP Detection

Prevents a student physically present in the room from hotspotting or scanning on behalf of peers who gave them their login credentials.

If requests from distinct student accounts originate from the same IP address within a **2.0-second burst window**, they are flagged as a `DUPLICATE_IP` anomaly.

Tier 4 Hardware Device Fingerprinting

Prevents proxying achieved by logging out and logging into a friend's student account on the same physical phone.

The system checks whether the same browser signature and user-agent string have already been used to mark attendance for another student in the same session.

```
// Haversine Great-Circle Distance Formulation: Δlat = toRadians(lat_session - lat_student); Δlng = toRadians(lng_session - lng_student); a = sin²(Δlat / 2) + cos(toRadians(lat_student)) * cos(toRadians(lat_session)) * sin²(Δlng / 2); c = 2 * atan2(√a, √(1 - a)); distanceInMeters = 6371000 * c; // Anti-Proxy Rule Enforcement: if (distanceInMeters > session.radiusMeters) { flagQuarantineViolation("GEOLOCATION_MISMATCH", distanceInMeters); }
```

4. Core Functional Modules & User Role Portals

4.1 Institutional Administrator Portal

- **Executive Metrics:** Real-time tally of enrolled students, active faculty, overall campus attendance rate, active batches, and unresolved security flags.
- **Interactive Analytics (Recharts):** 14-day chronological attendance vs. proxy violation trend lines, department turnout bar charts, and attendance risk breakdown donuts.
- **Anti-Proxy Threat Quarantine Center:** Real-time security feed showing flagged students, reasons (e.g., *"Student is 1,420m away"*), and 1-click administrative actions: `APPROVE` (manual override to Present), `BLOCK` (confirm Absent), or `FLAG`.
- **Immutable System Audit Logs:** Cryptographically logged ledger tracking all manual faculty status modifications, medical exemptions, and admin interventions.

- **Curriculum Management:** Full CRUD operations for creating subjects, assigning faculty leads, and managing student course enrollments.

4.2 Faculty Lecture Terminal

- **Active Session Launcher:** Allows professors to select a course, set the session type (`IN_PERSON`, `HYBRID`, or `ONLINE`), and adjust the geofence slider (20m to 200m).
- **Classroom QR Projector:** Large-screen dynamic QR projector with real-time countdown timer animation. Automatically handles token refresh cycles via Socket.io.
- **Live Attendee Stream:** Real-time feed of arriving students displaying avatars, names, roll numbers, timestamps, and verification methods without manual page refresh.
- **Manual Status Overrides:** Quick modal allowing professors to mark excused medical leaves or late arrivals.
- **Automated 75% Risk Roster:** Auto-identifies students whose attendance has breached the mandatory 75% limit and enables 1-click generation of formal parent warning notices.

4.3 Student Mobile Web Portal

- **Turnout Visualizer:** Circular aggregate progress meter and subject-by-subject attendance progress bars color-coded by compliance standing.
- **Active Lecture Banners:** Live pulsing notifications alerting the student whenever an enrolled course launches an active roll-call session.
- **"Scan & Attend" Camera Interface:** Integrated HTML5 camera reader utilizing `html5-qrcode` that operates directly in mobile Chrome and Safari without native app installation.
- **AI Face Verification Terminal:** WebRTC camera stream rendered onto HTML5 Canvas with simulated bounding-box landmark tracking and neural feature alignment (99.4% confidence score).
- **Personal Attendance Ledger:** Detailed chronological log of all historical sessions, timestamps, methods (`QR_CODE`, `FACIAL_REC`, `MANUAL`), and statuses.

5. Entity-Relationship Data Model & Schema

The persistence layer is structured using PostgreSQL 18 with pure JavaScript Prisma ORM models:

ENTITY	KEY FIELDS & TYPES	FUNCTIONAL PURPOSE
`User`	`id` (UUID), `email` (Unique), `name`, `passwordHash`, `role` (ADMIN, FACULTY, STUDENT), `department`, `rollNumber`	Central identity entity with role-based access controls and academic department linkage.
`Course`	`id` (UUID), `code` (Unique), `name`, `department`, `facultyId` (FK → User)	Academic subjects belonging to specific departments with assigned faculty instructors.
`Enrollment`	`id` (UUID), `studentId` (FK → User), `courseId` (FK → Course)	Many-to-many relationship defining authorized students for a given course.
`LectureSession`	`id` (UUID), `courseId` (FK), `startTime`, `endTime`, `sessionType`, `activeQrToken`, `latitude`, `longitude`, `radiusMeters`	Individual lecture instances. Stores the currently active rolling cryptographic token and geofence coordinates.
`AttendanceRecord`	`id` (UUID), `sessionId` (FK), `studentId` (FK), `timestamp`, `status` (PRESENT, ABSENT, LATE, FLAGGED_PROXY), `method`, `ipAddress`, `deviceInfo`	Immutable record of a student's verification event, capturing physical IP, device signature, and verification method.
`ProxyAlert`	`id` (UUID), `sessionId` (FK), `studentId` (FK), `reason`, `flaggedAt`, `resolved` (Boolean)	Quarantine record generated whenever an anti-proxy rule (geofence, IP, token age) is violated.

6. RESTful API & WebSocket Specification

METHOD	ENDPOINT PATH	RBAC ACCESS	OPERATION & PAYLOAD SUMMARY
`POST`	`/api/auth/login`	Public	Authenticates email/password, issues JWT with user role and profile.
`GET`	`/api/admin/dashboard`	`ADMIN`	Aggregates campus metrics, 14-day trend series, department turnout, and recent proxy flags.
`PATCH`	`/api/admin/proxy-alerts/:id`	`ADMIN`	Takes triage action: `APPROVE` (overrides to Present), `BLOCK`, or `FLAG`.
`GET`	`/api/admin/audit-logs`	`ADMIN`	Returns immutable chronological log of all manual overrides and exemptions.
`POST`	`/api/faculty/sessions`	`FACULTY`, `ADMIN`	Creates a new lecture session, initiates rolling QR generation, and joins socket room.
`GET`	`/api/faculty/sessions/:id/live`	`FACULTY`, `ADMIN`	Returns live session statistics, enrolled count, check-in records, and current QR token.
`POST`	`/api/student/scan-qr`	`STUDENT`, `ADMIN`	Submits `{ qrToken, latitude, longitude, deviceInfo }` for 4-tier anti-proxy validation.
`POST`	`/api/student/face-verify`	`STUDENT`, `ADMIN`	Verifies WebRTC canvas snapshot, confirms biometric liveness, and marks record as PRESENT.
`GET`	`/api/student/dashboard`	`STUDENT`, `ADMIN`	Returns student's per-course attendance breakdown, active class alert banners, and recent logs.
`POST`	`/api/attendance/generate-alert`	`FACULTY`, `ADMIN`	Generates formal administrative warning letters for students below 75% attendance.

7. Demonstration Credentials & Quick Reference

The system includes a pre-seeded dataset containing 15 students across 3 departments, 6 academic courses, 20 past lecture sessions, and 160+ historical attendance records:

Admin Account

Email: `admin@edupulse.com`
Password: `admin123`
Role: Institutional Dean & Audit Lead

Faculty Lead

Email: `faculty1@edupulse.com`
Password: `faculty123`
Role: CS301 Course Instructor (Live Class)

Student Account

Email: `student1@edupulse.com`
Password: `student123`
Role: Enrolled CS301 Student Scanner

Summary of Architectural Guarantee

> EduPulse eliminates the fundamental vulnerability of static attendance tracking. By combining dynamic cryptographic tokens refreshed every 7 seconds via WebSockets with physical GPS geofencing, duplicate IP network anomaly analysis, and browser-native camera verification, the platform delivers zero-hardware-fixture, proxy-proof educational data management.